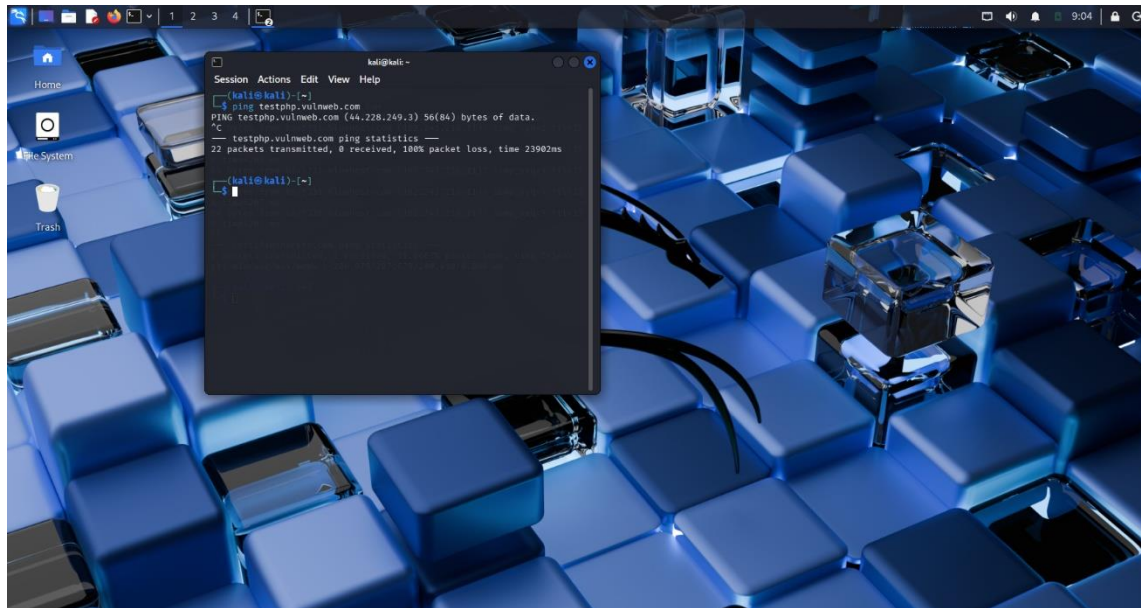


SCANNING NETWORKS

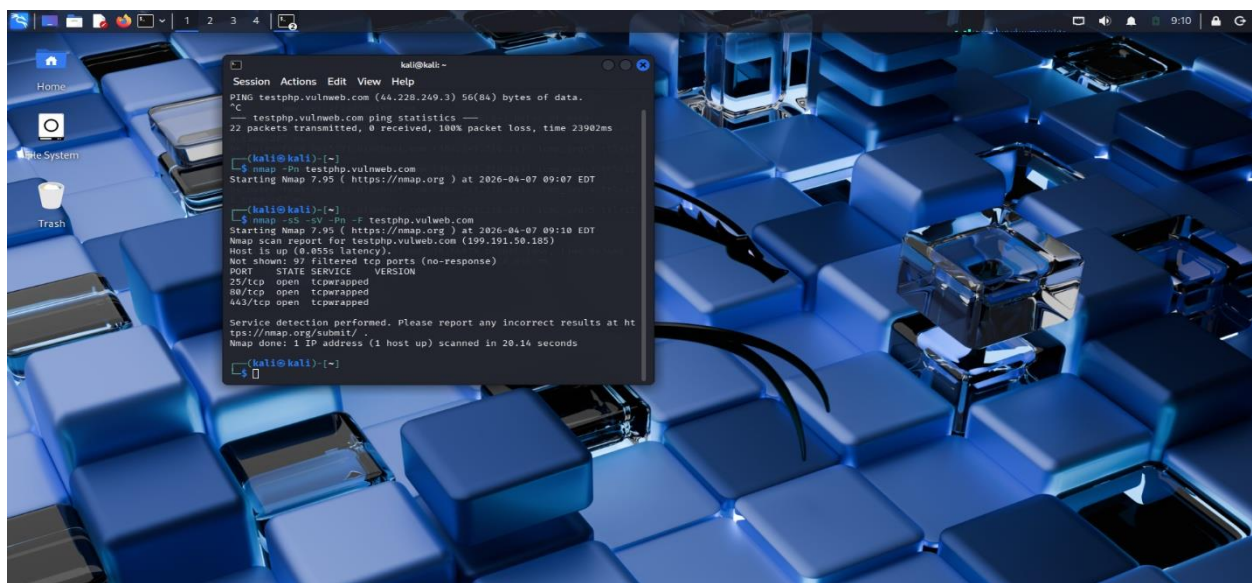
Task 1:

Ping was used to determine whether the host is up and running, but there were no responses (100% packet loss). This probably indicates that the server denies ICMP queries for security purposes rather than being down.



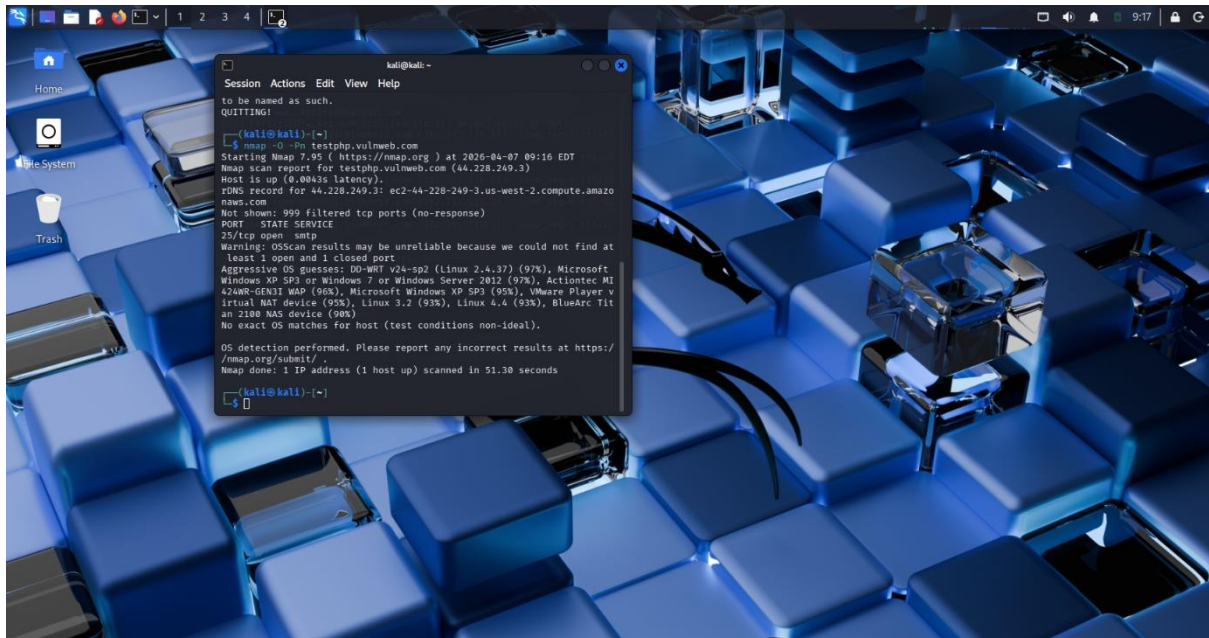
Task 2:

Using an Nmap scan, the host was found to be up with the following ports open: 25 (SMTP), 80 (HTTP), and 443 (HTTPS). All other ports have been blocked by a firewall, while the services are also 'tcpwrapp'.



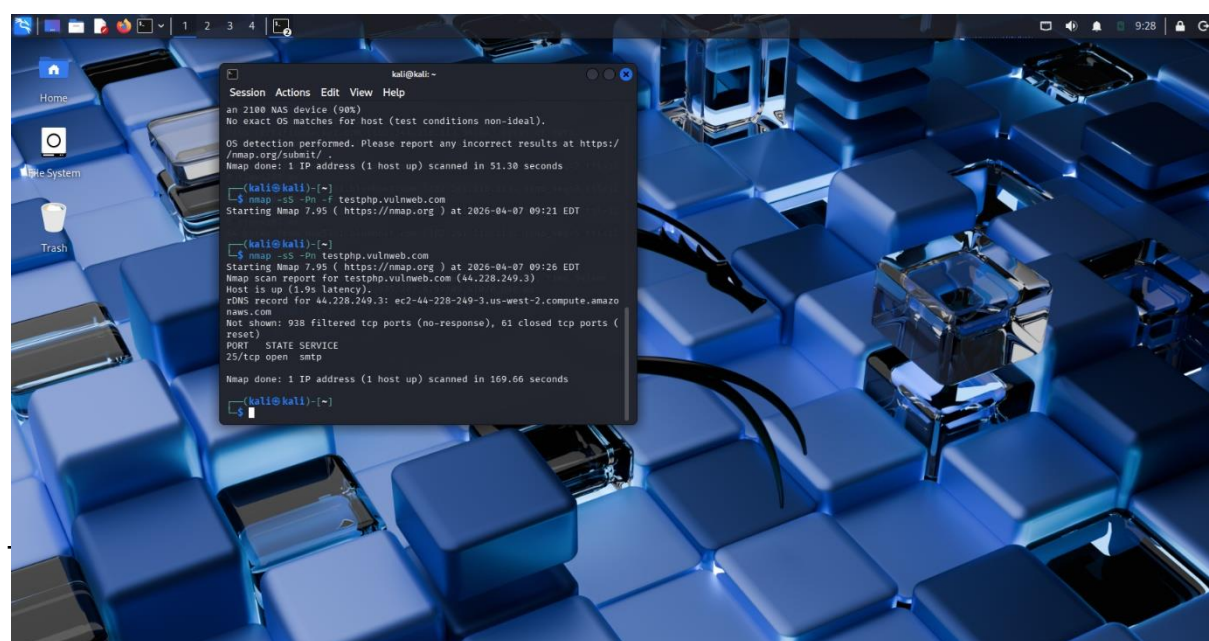
Task 3:

OS detection from Nmap did not provide the specific system information, indicating many possibilities for guessing. Firewalls or intrusion detection systems must have filtered out fingerprinting attempts, indicating proper security measures against OS detection

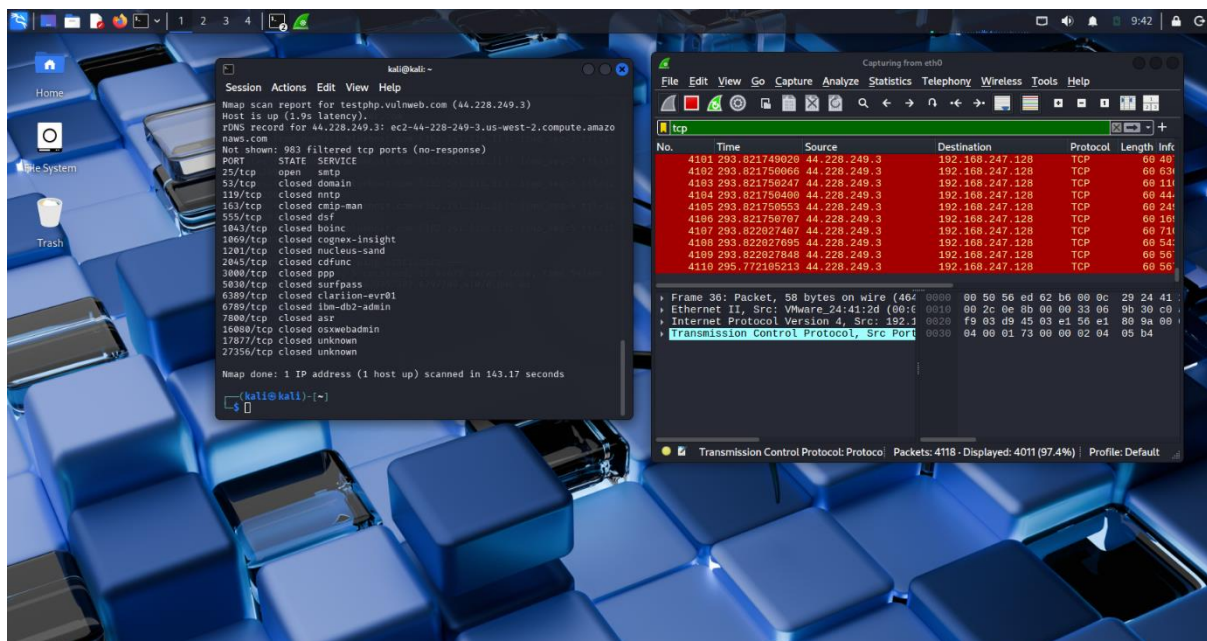


Task 4:

The Nmap OS detection (-O) could not pinpoint the exact operating system, presenting several guesses such as Linux and Windows. This is attributed to firewalls/IDS blocking fingerprinting, implying that stringent security protocols prevent OS detection.



TCP traffic was analyzed using Wireshark when an Nmap scan was performed. The result includes SYNs, SYN/ACKs, and RSTs, which show whether there were open ports or not.



Task 6:

Network security testing for a target system was performed by means of ping, Nmap, and Wireshark. It was impossible to use the ping technique since no answer to the command was provided, and thus there was ICMP blocking.

The detailed analysis of the ports made through the utilization of Nmap showed that the system has open ports, like 25, 80, and 443. At the same time, some of the ports are filtered, which indicates firewall protection.

Service detection showed that the ports used were under protection through tcpwrapped, so it became impossible to obtain much information on these ports. OS detection could not help because of restricted responses.

Packet Fragmentation revealed that firewall behaves differently according to different scanning methods, which means the usage of an intrusion detection system.

Wireshark also helped analyze network scanning by displaying TCP packet flows such as SYN and SYN-ACK packets.

In conclusion, the system is highly protected since firewall is used in addition to ICMP blocking. Nonetheless, there are open ports such as those for SMTP and web service that can be considered potentially hazardous.

Recommendations

To enhance security:

- Close unnecessary ports
- Update services regularly
- Secure firewall rules
- Constantly monitor network traffic